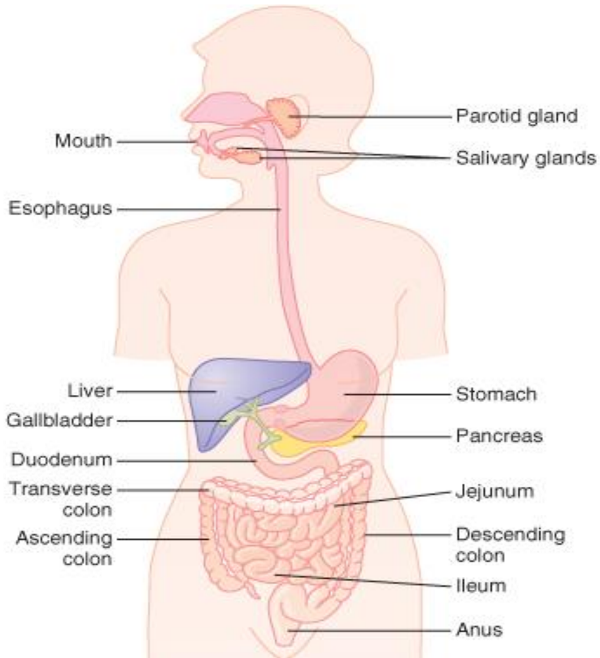
The Digestive System

The digestive system consists of:

- ***A.*** Gastro intestinal tract (GIT) which includes mouth, pharynx, oesophagus, stomach, intestine and anal canal. It is about 4.5 : 5 meters in length.
- ***B.*** Digestive glands that secrets digestive enzymes into the GIT. They include salivary glands, liver and pancreas.



Functions of the digestive system:

1- *Digestion:* means breakdown of large molecules to small absorbable molecules.

2- *Absorption:* means passage of the digested food to the blood via crossing the mucosa of the GIT.

3- *Secretion:* means secretion of saliva and other GIT juices.

4- *Motility:* which helps transport of food from the mouth down to all parts of the alimentary canal.

Salivary glands

Saliva is secreted by 3 pairs of salivary glands:

1- Parotid glands.

2- Submandibular glands.

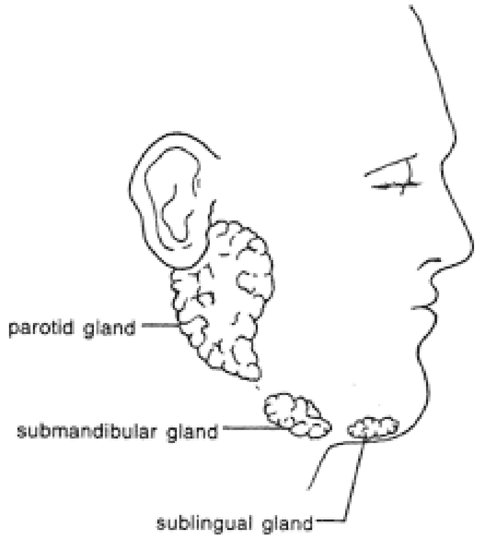
3- Sublingual glands.

- The daily secretion of saliva ranges between 1.5 – 2 L/day.
- pH: 6.8 (slightly acidic).

1-Parotid glands: produce a serous, watery secretion (25%).

2-Submandibular glands: produce a mixed serous and mucous secretion (70%).

3-Sublingual glands: secrete mainly mucous secretion (5%).



<u>Composition of saliva:</u>

It contains H_2O: 99.5 % and Solids: 0.5 %. (Organic and inorganic).

Organic compounds: mucous, salivary amylase, glucose, lingual lipase and lysozomes.

Inorganic compounds: Na salts, K salts and Ca phosphate.

<u>Functions of saliva</u>

<u>1- Digestive function:</u>

- Salivary amylase starts digestion of starch.
- Lingual lipase helps lipid hydrolysis in the stomach.

<u>2- Excretory function:</u>

It can excrete waste products as urea, iodides and fluorides to protect teeth enamel and prevents dental caries.

<u>3- Buffering action:</u>

It contains bicarbonate and phosphate buffers that helps to keep the pH of the mouth at 7 that keep the Ca in teeth and prevents its loss, so teeth remains healthy and strong.

<u>4- Cleansing action:</u>

It cleans the mouth & teeth by continuous flow and has antibacterial action

<u>Control of salivary secretion:</u>

Secretion of saliva is under control of the autonomic nervous system:

- ***Parasympathetic stimulation*** produces a large volume of saliva that is rich in electrolytes.
- ***Sympathetic stimulation*** produces a small, thick secretion that is rich in enzyme.

Salivary secretion has 2 phases:

A. *Conditioned(Aquired) reflex:* starts before food enters the mouth by sight, smell, or thinking of food .

B. *Simple (Inborn)(Unconditioned) reflex:* the presence of food or any object in the mouth stimulates salivary secretion.

Conditioned reflex:

1-Specific visual, auditory, olfactory or memory stimuli associated with food, consciously interpreted in cerebrum.

2- Impulses sent to salivary centre in medulla.

3-Saliva production initiated,

Simple reflex:

1- Food ingestion stimulates both mechanical & chemical receptors in the mouth.

2-Sensory impulses sent to salivary centre in medulla.

3-Autonomic impulses from medulla initiate saliva production.

Abnormalities of Salivary Secretion:

1-Excessive salivation is a symptom of almost any lesion in the oral cavity.

2-Xerostomia: decreased salivary secretion, may lead to oral infections

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The Stomach

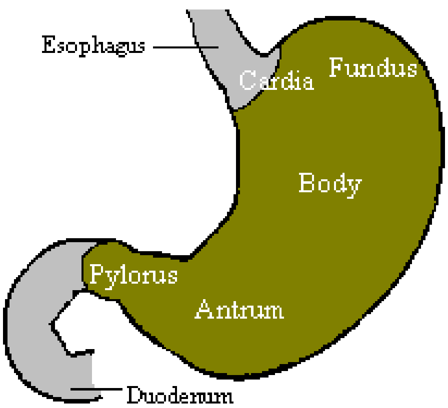
- The main parts are:

A. Fundus:

B. Body

C. Antrum

D. Pylorus



- The food enters the stomach in a semisolid form (*bolus*), is turned into a semi-liquid form by the *grinding action* of the stomach and the gastric juice.
- As food is liquefied in the stomach it passes through the pyloric canal into the small intestine.

Functions of the stomach:

1. It serves as a short-term storage reservoir.
2. Grinds and mixes food with gastric secretions, resulting in ***liquefaction*** of food**.**
3. Slowly releases food into the small intestine for further processing.
4. Starts digestion of some foods particularly proteins.
5. Releases iron from its compounds

Gastric Secretion:

*About 2.5 liters are secreted into the lumen of the stomach daily (**gastric juice**):*

The gastric juice consists of:

1-Mucus: a bicarbonate-rich mucus that coats and lubricates the gastric surface to protect it from acid and other agents.

2-Acid: hydrochloric acid is secreted from *parietal cells* into the lumen providing an acidic medium (pH: 1-2 in the lumen).

- This acid is important for activation of *pepsinogen* and inactivation of ingested microorganisms such as bacteria.

3-Pepsinogen: an inactive protease is secreted into gastric juice from the chief cells.

- Once secreted, pepsinogen is activated by Hcl acid into the active protease pepsin.
- Pepsin initiates digestion of proteins.

4-Intrinsic factor: secreted by *parietal cells* that is necessary for intestinal absorption of vitamin B_{12}.

5-A number of other enzymes are secreted by gastric epithelial cells e.g ***lipase***.

N.B.

In addition to the gastric juice the stomach secretes hormones (into the blood): *eg.* ***gastrin****, a peptide that is important in control of acid secretion and gastric motility.*

Control of gastric secretion:

- It is divided into 3 phases:

1-Cephalic phase:

...seeing, smelling and anticipating food is perceived in the brain.

A. The brain informs the stomach that it should prepare for reception of a meal.
B. This communication is composed of *parasympathetic stimuli* transmitted thought the vagus nerve.
C. The *vagus nerve stimulation* releases Ach and stimulates gastrin and histamine secretion.
D. Ach, gastrin and histamine stimulate secretion of gastric juice.

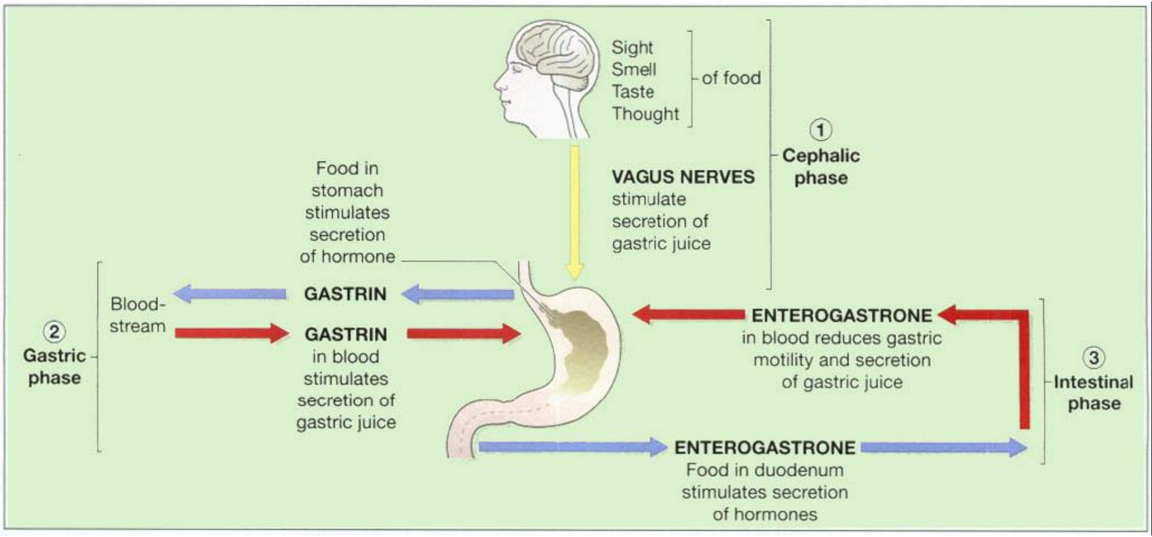
2-Gastric phase:

- When a meal enters the stomach ,it causes distension and mucosal irritation.
- These stimulate further secretion of gastric juice, by *increasing gastrin & histamin, more.*

3-Intestinal (inhibitory) phase:

- The wall of the small intestine has chemical and osmoreceptors
- If the contents reaching the small intestine are highly acidic or hyperosmolar, the small intestine sends inhibitory impulses and inhibitory hormones to the stomach.
- The impulses and hormones are conveyed to the stomach to inhibit it's secretion.

The following shape showing Control of gastric secretion:

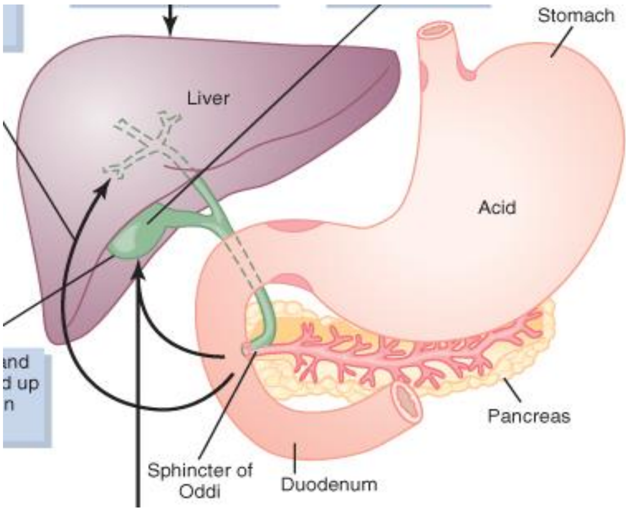


N.B.

Two of the hormones {secretin and cholecystokinin (CCK)} are forming enterogastrone.

The pancreas

Pancreas is considered as an exocrine and endocrine gland. The exocrine part secretes the pancreatic juice. The endocrine part secretes hormones like insulin.



Composition of pancreatic juice

- It is an alkaline secretion: the pH is 8.
- It contains H_2O : 98.5 %

Organic compounds: Pancreatic enzymes, trypsin inhibitor and co-lipase.

Inorganic compounds: Na bicarbonate mainly, K^+, Mg^{++} , Cl^- and Ca^{++}.

Functions of pancreatic juice

1- Na bicarbonate:

- Neutralizes the acids which come from the stomach to the duodenum and this:
-Prevents development of duodenal ulcer.
-Adjusts the pH for the action of pancreatic enzymes.

2- Digestion of protein:

- By trypsin, chymotrypsin, carboxypeptidase enzymes.

3- Digestion of lipids:

- By *pancreatic lipase* it acts on triglycerides to form monoglycerides and fatty acids.
- *phospholipase A_2.*

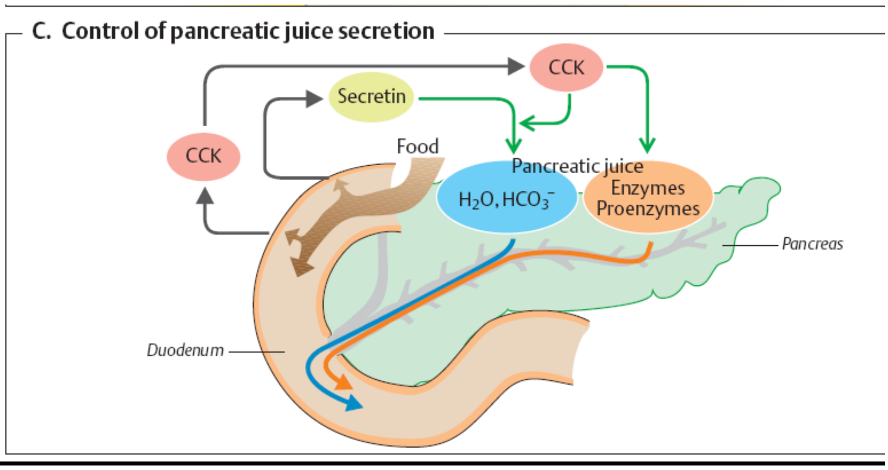
4- Digestion of carbohydrate:

By *pancreatic amylase* it converts starch, glycogen to disaccharides.

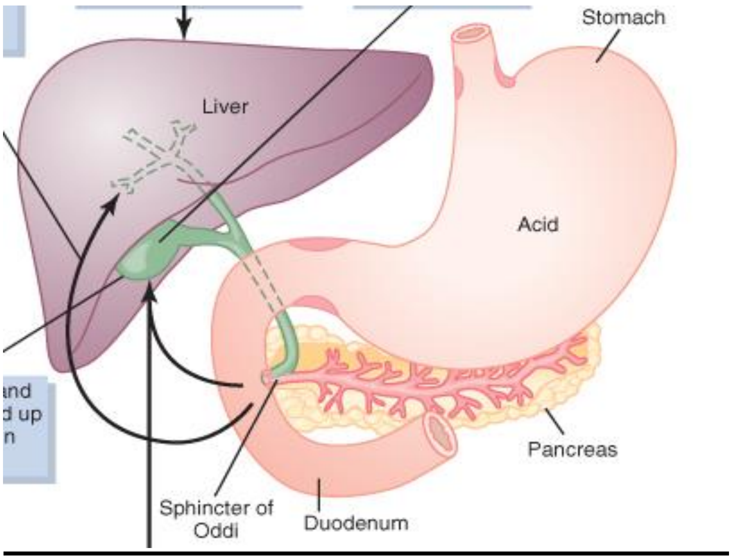
Control of Pancreatic Secretion:

Pancreatic secretion is stimulated by:

A) Secretin: *produced by the duodenum, stimulates HCO_3 secretion.*
B) Cholecystokinin(CCK): *produced by the duodenum stimulates enzyme secretion.*
***C)* Parasympathetic (Vagus):** *stimulates enzyme secretion.*



The liver



Functions of the liver

1- Metabolic functions:

A. *Carbohydrate metabolism:* it preserves the level of glucose constant, also it stores large amount of glycogen up to the time of need.
B. *Fat metabolism:* it helps oxidation of fatty acids to supply energy. Also, helps lipid formation from protein and carbohydrates.
C. *Protein metabolism:* needed for plasma protein formation and synthesis of urea from amonia.

2- Excretory functions:

- Liver excretes bile which is golden yellow secretion that contains bile salts which:

a. Help emulsification and digestion of lipids.
b. Absorption of fat and fat soluble vitamins.
c. Antiputrifactive function, because in absence of bile the undigested fat envelops protein particles prevents their digestion so intestinal bacteria attacks it causing its putrefaction.

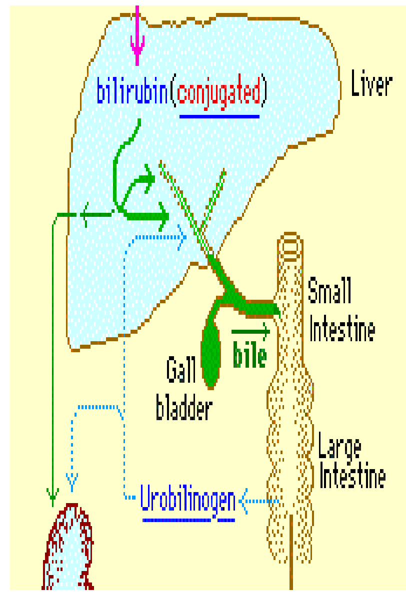
3- Detoxification functions:

Several toxic drugs or other toxic substances are detoxified in the liver.

THE DIGESTIVE SYSTEM

Bile

- Bile is a yellow secretion formed by the liver cells.
- It is collected by the bile ducts and secreted into the duodenum at the time of meals.
- In between meals bile is stored in the *gall bladder*.



Function:

A) Bile salts are responsible for *emulsification* of fat.

B) Excretion of bilirubin (product of hemoglobin degradation).

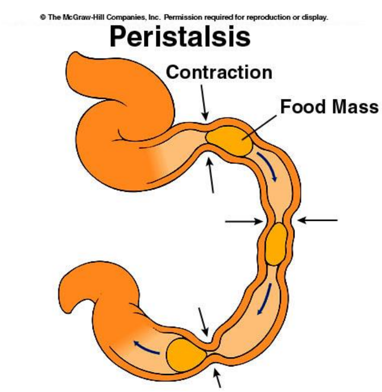
C) Bile salts make fatty acids soluble, enabling both these and fat-soluble vitamins (e.g. vitamin K) to be readily absorbed.

D) Stercobilin colors and deodorises the faeces.

Control of Bile Secretion:

- Bile secretion is stimulated by the *vagus nerve* and the hormone *cholecystokinin* (***CCK)***.
- The secretion starts when a meal enters the duodenum.
- Obstruction in the bile ducts e.g by stone causes *jaundice* due to increased bile pigments in blood.

Motility (movement of the stomach and intestine):

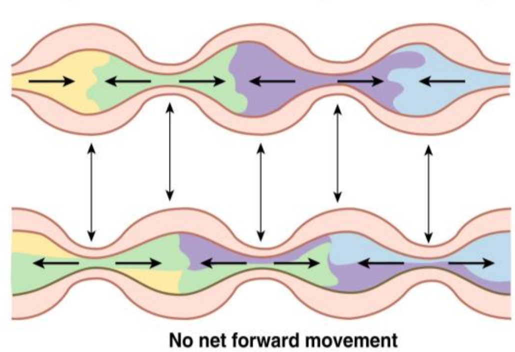


2 Main types of motility:

1-Propulsive movement (Peristalsis):

- carrying the food down the GIT at varying speeds, with the rate of propulsion
- dependent on the functions of different regions
- It is stimulated by distension.
- The part of the intestine behind the food contracts and the part in front of the food dilate.

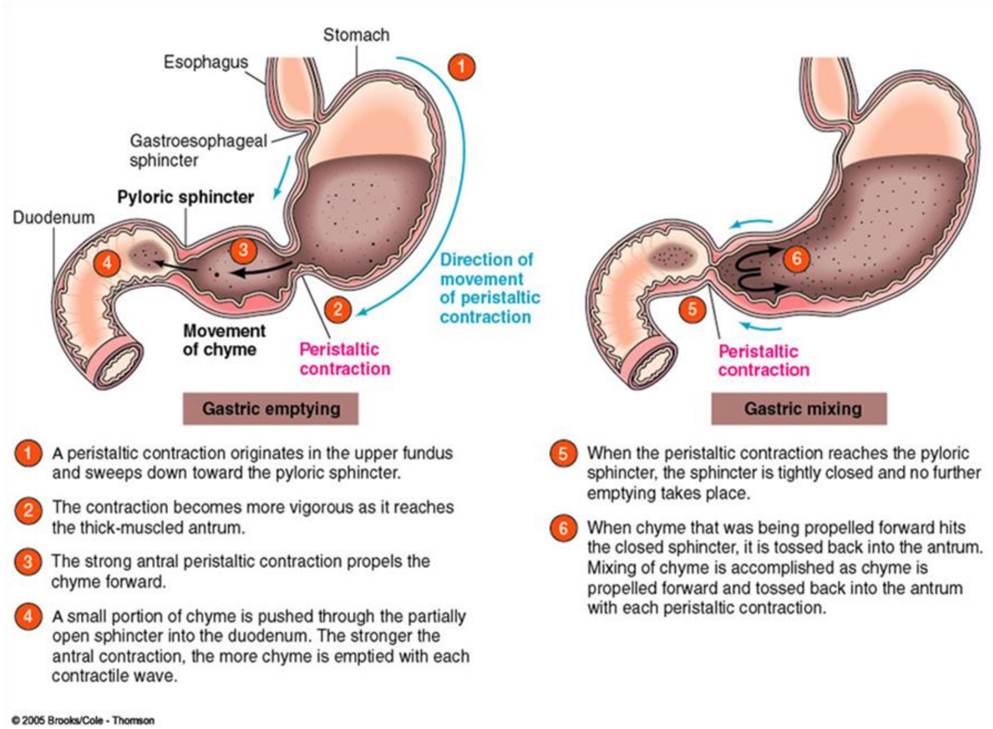
(b) Segmental contractions are responsible for mixing.



2-Mixing-Movement (Segmentation-Movement):

a. It mixes food with digestive juices.

b. facilitate absorption by increasing interactions between contents and surfaces of tract



<u>Patterns of GI motility:</u>

Type of contraction	***Organ/structure***
• *Tonic contractions*	upper and lower esophageal sphincters pyloric valve sphincter of Oddi ileocecal valve internal anal sphincter
• *Propulsive peristalsis*	esophagus lower 2 thirds of stomach small intestine rectum
• *Reverse peristalsis (antipropulsion)*	proximal colon
• *Mass movements*	ascending, transverse and descending colon
• *Nonpropulsive segmentation*	small intestine

• *Haustration* ascending, transverse and descending colon

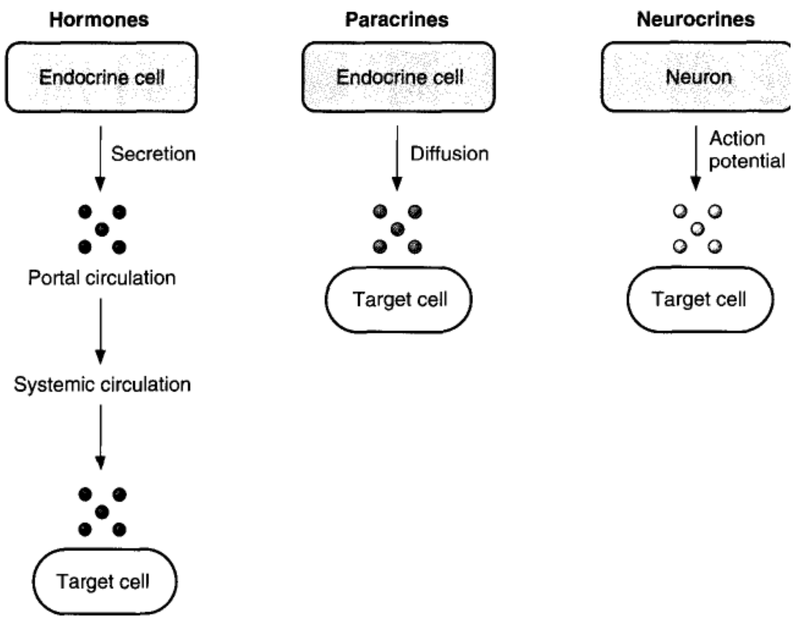
Movement of materials along the digestive tract is controlled by:

1- ***Neural mechanisms***
   - Parasympathetic and local reflexes

2- ***Hormonal mechanisms***
   - Enhance or inhibit smooth muscle contraction

3- ***Local mechanisms***
   - Coordinate response to changes in pH or chemical stimuli

Gastrointestinal hormones, paracrines, and neurocrines:



A- Gastrointestinal hormones:

1. Are released from endocrine cells in the GI mucosa into the portal circulation, enter the general circulation, and have physiologic actions on target cells. These hormones include the following hormones.

<u>1- Gastrin</u>

- Secret from G cells of stomach

Actions of gastrin:

- H^+ secretion by the gastric *parietal cells.*
- Stimulates growth of gastric mucosa

Stimuli for secretion of gastrin:

- Small peptides and amino acids in the lumen of the stomach
- Distention of the stomach

- Vagal stimulation via gastrin-releasing peptide (GRP)

Inhibition of gastrin secretion:

- Somatostatin
- H^+ in the stomach (negative feedback control)

2- *CCK*

2. is released from the ***I cells*** of the duodenal and jejunal mucosa

***Actions of CCK*:**

- Stimulates *contraction of the gallbladder* and *relaxation of the sphincter* of Oddi for secretion of bile.
- Increase pancreatic enzyme secretion and HCO_{3^-} secretion.
- Increase growth of the exocrine pancreas.
- Inhibits gastric emptying.

Stimuli for the release of CCK

- Small peptides and amino acids
- Fatty acids and monoglycerides

3-Secretin

3. is released by the ***S cells*** of the duodenum

Actions of secretin:

- Are coordinated to reduce the amount of H^+ in the lumen of the small intestine.
- Increase pancreatic HCO_3- secretion and increases growth of the exocrine pancreas.
- Increase HCO_3- and H_2O secretion by the liver, and increases bile production.
- Inhibits gastric H^+ secretion.

Stimuli for the release of secretin:

- H^+ in the lumen of the duodenum
- Fatty acids in the lumen of the duodenum

4-GIP (glucose-dependent insulinotropic peptide)

- is secreted by the duodenum and jejunum

Actions of GIP:

- Increase insulin secretion.
- Inhibits gastric H^+ secretion by parietal cells.

Stimuli for the release of GIP:

4. is stimulated by fatty acids, amino acids, and orally administered glucose.

B- Paracrines

5. Are released from endocrine cells in the GI mucosa.
6. The GI paracrines are *somatostatin and histamine.*

1- Somatostatin

- is secreted by cells throughout the GI tract in response to H^+ in the lumen.
- Its secretion is inhibited by vagal stimulation.

Action of Somatostatin:

- inhibits the release of all GI hormones.
- inhibits gastric H^+ secretion.

2- Histamine

- is secreted by mast cells of the gastric mucosa.
- increases gastric H^+ secretion

C- Neurocrines

- Are synthesized in neurons of the GI tract, and released by action potentials in the nerves.
- The GI neurocrines are vasoactive intestinal peptide (VIP), GRP (bombesin), and enkephalins.

1- VIP

- Is released from neurons in the mucosa and smooth muscle of the GI tract.
- Produces relaxation of GI smooth muscle, including the lower esophageal sphincter.
- Increase pancreatic HCO_3- secretion and inhibits gastric H^+ secretion.
- In these actions, it resembles secretin.

2- GRP (bombesin)

- Is released from vagus nerves that innervate the G cells.
- Increase gastrin secretion from G cells.

3- Enkephalins

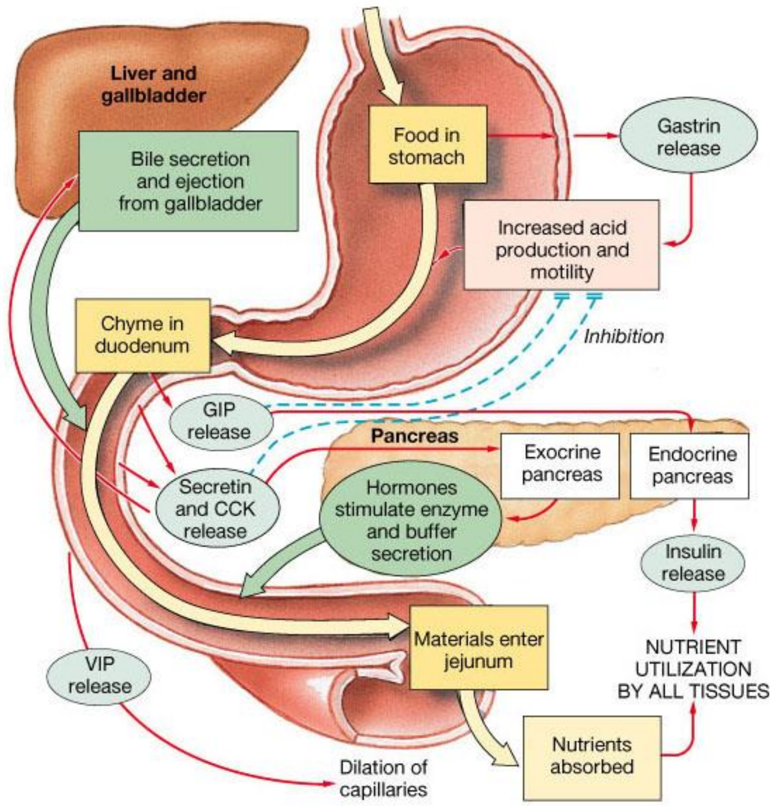
- are secreted from nerves in the mucosa and smooth muscle of the GI tract.
- Stimulate contraction of GI smooth muscle, particularly the lower eso-phageal, pyloric, and ileocecal sphincters.
- Inhibit intestinal secretion of fluid and electrolytes. This action forms the basis for the usefulness of opiates in the treatment of diarrhea.

Additional GI hormones

- Hormones are produced by enteroendocrine cells in the GI tract in stomach, small and large intestine

Motilin	increases intestinal motility
Serotonin	increases intestinal motility
Neurotensin	decreases intestinal motility
Glucagon	increases blood flow to ileum
Entero-glucagon	stimulate hepatic glycogenolysis
enterogastrone.	reduces gastric motility and secretion of gastric juice
Urogastrone (Epidermal Growth Factor)	inhibits secretion of HCl increases epithelial growth

The Activities of Major Digestive Tract Hormones



The Small Intestine:

- The small intestine is the longest section of the digestive tube extending from the pylorus to the large intestine.

It consists of 3 segments:

- *Duodenum*
- *Jejunum*
- *Ileum*

- The small intestine is the site for digestion and absorption of almost all nutrients.
- The small intestine is essential for life.

Functions of the small intestine:

1. Onward movement of its contents which is produced by peristalsis.
2. Secretion of intestinal juice.
3. Completion of chemical digestion of carbohydrates, protein and fats in the enterocytes of the villi.
4. Protection against infection by microbes.
5. Secretion of the hormones cholecystokinin (CCK) and secretin.
6. Absorption of nutrients.

Digestion in Small Intestine:

A) Once within the small intestine, the food particles are exposed to *pancreatic enzymes , bile, and intestinal juice* which enables digestion of all food particles.

B) The final stages of digestion occur on the surface of the small intestinal epithelium (the brush border).

- Carbohydrates are broken down to monosaccharides
- Proteins are broken down to amino acids
- Fats are broken down to fatty acids and glycerol.

Absorption in Small Intestine:

- The absorptive surface area of the small intestine (the brush border) is about 250 square meters .
- This very large surface area is because the wall of the small intestine *has 3 modifications:*

A- <u>*Mucosal folds*:</u> the inner surface of the small intestine is not flat, but thrown into circular folds.

B- <u>*Villi:*</u> the mucosa forms projections which protrude into the lumen .

C- <u>*Microvilli:*</u> the villi is studded with densely-packed microvilli.

Absorption in the Small Intestine:

7. All nutrients from the diet are absorbed into blood across the mucosa of the small intestine.
8. In addition, the intestine absorbs water and electrolytes

Absorption of Water and Electrolytes:

- The small intestine absorbs massive quantities of water:
- A normal person ingests roughly 1 to 2 liters of dietary fluid every day. On top of that, another 6 to 7 liters of fluid is received by the small intestine daily as

secretions from salivary glands, stomach, pancreas, liver and the small intestine itself.

- By the time the contents enters the large intestine, approximately 80% of this fluid has been absorbed.
- The absorption of water is dependent on absorption of solutes, particularly sodium.
- Interference with water absorption may cause diarrhoea.

Digestion and absorption of carbohydrates:

i) Polysaccharides and disaccharides :

- Must be digested to monosaccharides prior to absorption:

a) The main dietary carbohydrate is ***starch***. Digestion of starch starts in the mouth by the *salivary amylase*. *Pancreatic amylase* continues the digestion of carbohydrates. *Amylase* breaks down starch into *maltose*.

b) Disaccharides:

The dietary disaccharides, *lactose* (milk sugar) , *sucrose* (table sugar) , and *maltose* , are broken down by enzymes on the wall of the small intestine(the brush border). These enzymes include:

9. *Maltase* cleaves maltose into two molecules of glucose.
10. *Lactase* cleaves lactose into a glucose and a galactose.
11. *Sucrase* cleaves sucrose into a glucose and a fructose.

c)Monosaccharides:

- The monosacharides, *glucose*,*galactose* and *fructose* are absorbed into the intestinal cells and then into the blood.
- Deficiency of lactase may cause lactose intolerance (abdominal discomfort and diarrhea following intake of milk).

Digestion and absorption of proteins:

- They must be digested to amino acids first.
- Digestion of proteins starts in the *stomach.* The stomach secretes *pepsinogen* (activated to *pepsin* by the action of Hcl).
- The *pancreas* secretes a group of potent *proteases*, eg. *Trypsin and chymotrypsin.* They are secreted as inactive and activated after reaching the lumen of the intestine
- *Gastric and pancreatic proteases* digest dietary proteins within the lumen of the small intestine into medium and small peptides.

- The surface of the small intestine has *peptidase* enzymes. The peptidases complete the breakdown of peptides, converting them to free amino acids.
- The free amino acids are absorbed into the blood.

Digestion and Absorption of Lipids:

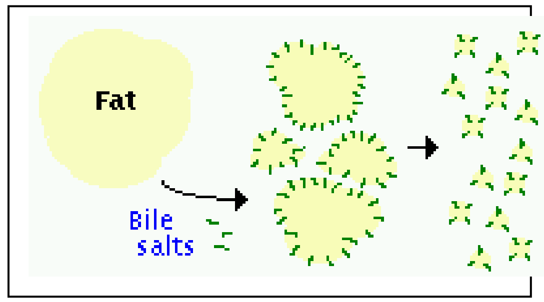
- The main dietary lipid is *triglyceride*, composed of a glycerol with 3 fatty acids. Additionally, most foodstuffs contain cholesterol, fat-soluble vitamins.
- In order for the triglyceride to be absorbed, *two processes* must occur:
  1- *Emulsification*: large aggregates of dietary triglyceride, which are insoluble in an aqueous environment, are broken down physically.
  2- *Digestion:* Triglyceride molecules must be enzymatically digested to produce monoglyceride and fatty acids.

Emulsification is done by bile salts.

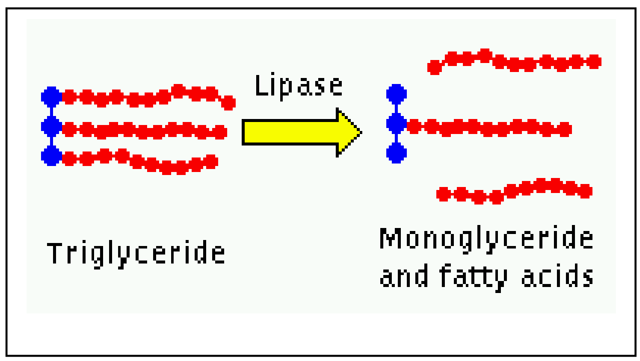
A- Bile salts have water soluble and fat soluble poles.

B- Lipids are dissolved by the fat soluble part (the inside).

C- The combinations of bile salts with various dietary lipids are called *micelles*.



- Digestion of triglyceride into monoglyceride and free fatty acids is accomplished predominantly by *pancreatic lipase*.



Micelles:

- Monoglycerides, fatty acids ,bile salts and other lipids form structures called ***micelles***
- Micelles have a hydrophilic surface

Chylomicrons:

- After entry into the intestinal cells, *lipids combine with proteins* to form chylomicrons inside the cells. The chylomicrons enter the lymphatics first before reaching the blood. Short chain fatty acids may enter the blood directly

Functions of the large intestine, rectum and anal canal:

1. ***Absorption***
   - Water until the faeces is semisolid
   - Mineral salts, vitamins and some drugs are also absorbed
2. ***Microbial activity***

- contain certain types of bacteria, which synthesize *vitamin K* and folic acid
- Gases in the bowel consist of N_2, O_2, CO_2, H_2 and methane is produced by bacterial fermentation of unabsorbed nutrients, especially carbohydrate.
- Gases pass out of the bowel *as flatus.*

3. ***Mass movement***

— About twice an hour does a wave of strong peristalsis sweep along the transverse colon forcing its contents into the descending and sigmoid colons.

4. ***Defaecation***

- When a mass movement forces the contents of the sigmoid colon into the rectum, the nerve endings in its walls are stimulated by stretch.
- The *external anal sphincter* is under conscious control through the *pudendal nerve.*
- defaecation involves *involuntary contraction* of the muscle of the rectum and *relaxation of the internal anal sphincter*.
- Contraction of the *abdominal* muscles and lowering of the *diaphragm* assist the process of defaecation